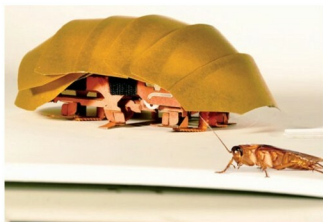


other GPUs. Hence, the cores would not have to transmit data back and forth the memory bank, thus saving time and energy. This also facilitates communication between individual cores without the need for it to be routed through the main memory.

Cockroach-like robot could save people during earthquakes

Scientists at University of California, Berkeley, USA, have drawn robotic inspiration in a cockroach's ability to squish through the tiniest of cracks and still move at tremendous pace even under compressed state.



CRAM, the compressible robot, with a real cockroach
(Image courtesy: www.cbsnews.com)

The flexibility of cockroaches allows them to travel at 4.8kmph. Using a high-speed camera, scientists observed that the insects could manage themselves into a space less than a quarter of their height by minimising the size of their exoskeletons to nearly half, and this adjustment would take them just a second. They also observed that the exoskeleton collapse does not slow them down; it continues by means of body-friction legged crawling.

This gave them a hidden potential robot. They laid out an arrangement of plates, laminated in a way similar to the exoskeleton and wrapped it around a robot. They named it CRAM, or Compressible Robot with Articulated Mechanisms.

Polymer-coated contact lenses transform into next-gen electronics

Scientists from Future Industries Institute (FII) of University of South Australia have completed research on polymer film coating, which can conduct electricity on a contact lens and possibly integrate electrical circuits that are safe for human wear. This breakthrough could provide one of the safest methods of bringing people and their smartdevices closer together, according to FII associate professor, Drew Evans.

The conductive polymer used in the study is poly (3,4-ethylenedioxythiophene) (PEDOT) deposited from vapour phase onto hydrated hydrogel substrates, which were further plasma-treated for providing better adhesive property.

Researchers also seek to produce medical benefits through biocompatible conducting polymers at nano-scale and grow these directly on a contact lens.

Bee pollen boosts battery performance

According to a new research, bee pollen could play an important role in battery research. Scientists at Purdue University, USA, have been researching ways to make better batteries, and have recently discovered that pollen grains and their unique microstructures could be put to use as an efficient type of energy-storage unit.

The scientists were trying to improve on conventional lithium-ion batteries, which have an anode made of carbon (usually graphite) and a cathode made of lithium-cobalt-oxide. The electrolyte that runs through the battery is made of lithium salts.

They found that if they could turn pollen into a carbon anode with a more useful microstructure than graphite, they might be able to create a battery with the ability to store more energy. So they took pollen from honeybees and cattails, and turned these into little pieces of carbon. They superheated a section of bee pollen and a section of cattail pollen to 1112°F (600°C) in a space that was filled with argon gas, which stops carbon from burning up.

They then reheated the pollen based carbon pieces to create more empty pockets in the pollen structure, which increases their capacity to store energy.

Traffic lights that recognise, favour cyclists

Forty per cent of residents in Copenhagen commute by bikes and have access to a city-wide network of protected bike lanes and sleek infrastructure built just for them. The city aims to be carbon-neutral by the year 2025.

To achieve this goal, the city's latest intelligent transport systems action plan calls for making riding a bike or taking the bus more appealing. Copenhagen wants to cut bus travel times by five per cent to 20 per cent and cycling travel times by ten per cent, among other measures. To do that, the city is installing 380 intelligent traffic signals that will spot and prioritise buses and bikes. This will ensure that traffic flows better, so people can save time in the greenest possible way.